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Symbiosis AI Course Session Plan

Revised for final-year engineering students

Preface

This document presents a structured 12-session teaching plan derived from the course Autonomous AI Systems and Agent-Based Computing. The revised version is tuned for final-year engineering students, so it moves quickly through fundamentals and spends more time on model behavior, retrieval design, agent workflows, evaluation, and real-world application patterns [file:1].

The intention is to convert the syllabus into a practical, classroom-ready format that is still rigorous enough for senior engineering students. Each session includes a focused lab-style exercise so students build, compare, analyze, or present something tangible rather than only submitting written work [file:1].

This plan maps each session to the course objectives to ensure the curriculum is covered end to end. The course objectives emphasize understanding GenAI and LLMs, learning embeddings and RAG, designing prompts, building AI applications with APIs and open-source tools, and evaluating limitations, risks, ethics, and governance aspects of AI [file:1].

Course context

The curriculum is organized into five major units: introduction to AI, Generative AI, and Agentic AI; prompt engineering and GenAI application development; NLP essentials and RAG; agentic AI concepts and workflow automation; and agent frameworks, evaluation, and applications [file:1].

For final-year engineering students, the same syllabus can be taught with a more advanced lens: less time on elementary definitions, more time on architecture, design trade-offs, retrieval quality, tool use, failure modes, and responsible deployment decisions [file:1].

The formal learning objectives still remain the guide: students should understand the fundamentals of Generative AI and large language models, embeddings, vector databases, and RAG systems; design effective prompts; build AI-powered applications using APIs and open-source tools; and evaluate limitations, risks, ethics, and governance aspects of AI [file:1].

Session plan table

Disclaimer: Lab exercises are proposed to make the course practical and demonstrable. They may be adjusted later based on student level, available tools, lab infrastructure, time constraints, or institutional requirements.

Session	Topic	Course Objective Mapping	Classroom focus	Lab exercise / deliverable
1	AI evolution, GenAI, and Agentic AI	CO1: Understand the fundamentals of Generative AI and large language models	AI evolution; traditional AI vs ML vs DL vs GenAI; what makes systems agentic	Create a one-slide architecture sketch showing a GenAI system vs an Agentic AI system for the same use case, such as resume generation or student support.
2	LLM fundamentals and behavior	CO1: Understand the fundamentals of Generative AI and large language models	Tokenization, context windows, inference behavior, temperature, top-p, and why models fail or drift	Run the same prompt with 3 different parameter or prompt variations and submit a comparison of output quality, consistency, and hallucination risk.
3	LLM use cases and model selection	CO1: Understand the fundamentals of Generative AI and large language models	Generation, summarization, classification, QA, and when to use prompting vs retrieval vs API orchestration	Compare two GenAI outputs for the same task and justify which one is better for a real-world use case based on structure, accuracy, and usability.
4	Prompt engineering as system design	CO2: Design effective prompts for different use cases	Role prompting, constraints, context injection, output schema, and prompt decomposition for complex tasks	Design a prompt set for a structured task like resume tailoring, meeting notes extraction, or job-description parsing, and show prompt evolution across iterations.
5	Structured outputs and prompt evaluation	CO2: Design effective prompts for different use cases	JSON, tables, templates, output reliability, prompt testing, failure modes, measurable improvement	Build a prompt that reliably outputs structured data, then test it on 3 different inputs and show where it succeeds or fails.
6	GenAI application development with APIs	CO2 and CO4: Design effective prompts; build AI-powered applications using APIs and open-source tools	System/user prompts, API integration, error handling, retry logic, rate limits, and basic app architecture	Build a small script or notebook that takes user input and generates a role-specific output such as a resume summary, email draft, or job-fit analysis.
7	NLP essentials, embeddings, and semantic search	CO3: Understand embeddings, vector databases, and RAG systems	Embeddings as vector representations, semantic similarity, retrieval, clustering, and search	Compare a resume against 3 job descriptions using embeddings or semantic scoring, and present which job is the closest fit with evidence.

8	RAG architecture and retrieval strategy	CO3: Understand embeddings, vector databases, and RAG systems	Chunking strategy, embedding choice, vector store behavior, top-k retrieval, context injection, failure cases	Build a mini-RAG pipeline over a small document set and demonstrate which chunks are retrieved for a few test questions.
9	RAG evaluation and grounded QA	CO3: Understand embeddings, vector databases, and RAG systems	Answer grounding, hallucination reduction, retrieval quality, and practical evaluation criteria	Create 5 document-based questions and evaluate whether the system answers from retrieved evidence correctly. Submit retrieved chunks plus final answers.
10	Agentic AI concepts and decision loops	CO4: Build AI-powered applications using APIs and open-source tools	Planning, reasoning, memory, tools, action, function calling, and how agents differ from automation	Design an agent workflow for a realistic task such as resume review, study planning, or job recommendation, showing tools, memory, and step sequence.
11	Multi-agent workflows and orchestration	CO4: Build AI-powered applications using APIs and open-source tools	Single-agent vs multi-agent designs, role specialization, orchestration, human-in-the-loop checks, control mechanisms	Create a multi-agent workflow where one agent drafts, one validates, one improves, and one finalizes an output. Present the workflow as a diagram.
12	Frameworks, evaluation, and responsible AI	CO5: Evaluate limitations, risks, ethics, and governance aspects of AI	LangChain, LlamaIndex, AutoGen, CrewAI, evaluation, reliability, bias, fairness, privacy, deployment concerns	Present a mini project proposal for an agentic AI application with architecture, chosen framework, evaluation metrics, risk analysis, and ethical safeguards.

Session 1

Introduction to AI, Generative AI, and Agentic AI

What will be covered

- Start with a rapid but clear recap of AI evolution: rule-based systems, machine learning, deep learning, generative models, and agentic systems [file:1].
- Focus on capability differences rather than just definitions, so students understand what changes when systems become generative or agentic [file:1].
- Use the resume-generation exercise as the concrete opening activity to show content generation as a practical use of GenAI [file:1].

Classroom / lab exercise

- Final-year emphasis: compare GenAI and Agentic AI as product patterns, not just buzzwords.
- Classroom lab: create a GenAI-generated resume for a fictional student profile and present the output.
- Ask students to briefly explain the prompt inputs they used and what changed in the output.

Session 2

LLM Fundamentals and Output Behavior

What will be covered

- Explain tokenization, transformer intuition, pretraining vs fine-tuning, context windows, inference, and parameter settings [file:1].
- Go deeper into output behavior: why the same model can produce different responses with changes in prompt wording or sampling settings.
- Use simple examples to show how context length and instruction quality affect model consistency and hallucination risk [file:1].

Classroom / lab exercise

- Final-year emphasis: treat this as model-behavior analysis, not just definitions.
- Classroom lab: run the same resume prompt with three variations and compare quality, completeness, and tone.
- Students should highlight where the model becomes more focused, more verbose, or less reliable.

Session 3

LLM Use Cases and Model Interaction

What will be covered

- Connect foundational LLM ideas to tasks such as generation, summarization, question answering, and model comparison [file:1].
- Discuss when an LLM can solve a task through prompting and when the application needs retrieval or orchestration.
- Introduce API interaction as a deployment pattern rather than a tutorial-level coding drill [file:1].

Classroom / lab exercise

- Final-year emphasis: use use-case selection and trade-offs as the main theme.
- Classroom lab: generate a resume summary or cover letter from the resume created earlier.
- Ask students to justify whether the output is better framed as generation, summarization, or transformation.

Session 4

Prompt Engineering Basics

What will be covered

- Treat prompt engineering as a design discipline: role prompting, context injection, constraints, and output shaping [file:1].
- Cover zero-shot, one-shot, and few-shot prompting in the context of reliable production behavior, not just experimentation.
- Use resume tailoring and structured extraction as the primary examples [file:1].

Classroom / lab exercise

- Final-year emphasis: prompts should be evaluated like specs, not like casual instructions.
- Classroom lab: create prompts that generate the same resume in professional, concise, and ATS-friendly styles.
- Have students compare the prompt patterns and explain why one style is more usable for a target role.

Session 5

Prompt Refinement and Structured Outputs

What will be covered

- Focus on iterative improvement, output formatting, and robustness across inputs [file:1].
- Show how to request JSON, markdown tables, templates, or other structured outputs for downstream systems.
- Introduce practical prompt testing by observing failure modes and refining instructions accordingly [file:1].

Classroom / lab exercise

- Final-year emphasis: emphasize repeatability and machine-readability of outputs.
- Classroom lab: improve a weak resume prompt step by step until it outputs a reliable structured format.
- Students should show before-and-after outputs and explain what change improved the result.

Session 6

GenAI Application Development with APIs

What will be covered

- Move from prompt design to application integration: system prompts, user prompts, API calls, rate limits, and error handling [file:1].
- Discuss how prompt logic becomes a reusable feature inside a larger application architecture.
- Keep the coding task practical but not overly long, so students focus on application flow and resilience [file:1].

Classroom / lab exercise

- Final-year emphasis: focus on architecture, reliability, and integration thinking.
- Classroom lab: build a notebook or script that takes user input and generates a role-specific output such as a resume summary, email draft, or job-fit analysis.
- Students should show the request flow, the prompt, and the result, plus note any handling for failed requests.

Session 7

NLP Essentials, Embeddings, and Semantic Search

What will be covered

- Explain text preprocessing, word embeddings vs contextual embeddings, and why semantic similarity is central to modern retrieval systems [file:1].
- Connect embeddings to practical tasks like search, matching, recommendation, and clustering.
- Make the link between embeddings and the coming RAG sessions very explicit [file:1].

Classroom / lab exercise

- Final-year emphasis: use embeddings as a system primitive, not as a vocabulary-only topic.
- Classroom lab: compare one resume against three job descriptions using embeddings or semantic scoring.
- Students should identify the closest match and justify the result using similarity evidence.

Session 8

RAG Architecture and Retrieval Strategy

What will be covered

- Explain why plain prompting is insufficient for document-grounded applications and how RAG solves the retrieval problem [file:1].
- Discuss chunking strategy, vector-store behavior, retrieval depth, and context injection trade-offs.
- Use resume/job-description retrieval as the working example for the session [file:1].

Classroom / lab exercise

- Final-year emphasis: focus on retrieval design choices and failure modes.
- Classroom lab: build a mini-RAG pipeline over a small document set and show which chunks are retrieved for test questions.
- Ask students to explain why the selected chunks are relevant or why retrieval might be poor.

Session 9

RAG Evaluation and Grounded QA

What will be covered

- Move from building RAG to evaluating it: grounding, hallucination reduction, answer correctness, and retrieval quality [file:1].
- Show that a good RAG system must answer from evidence, not just produce fluent text.
- Introduce practical evaluation criteria that students can apply in mini-projects [file:1].

Classroom / lab exercise

- Final-year emphasis: think like evaluators rather than just implementers.
- Classroom lab: create five document-based questions and test whether answers are supported by retrieved evidence.
- Students should submit retrieved chunks along with the final answer and a trustworthiness judgment.

Session 10

Agentic AI Concepts and Decision Loops

What will be covered

- Introduce agents as systems that plan, reason, remember, act, and use tools, going beyond pure generation [file:1].
- Explain task decomposition and function calling as the bridge between intelligence and action.
- Use resume review or job-fit analysis as the primary use case to show agent behavior in context [file:1].

Classroom / lab exercise

- Final-year emphasis: focus on control loops and decision structure.
- Classroom lab: design an agent workflow that reviews a resume against a job post and suggests improvements.
- Students should show the tools, memory points, and sequence of actions the agent would follow.

Session 11

Multi-Agent Workflows and Orchestration

What will be covered

- Compare single-agent and multi-agent systems and explain why role specialization can improve complex workflows [file:1].
- Cover orchestration, human-in-the-loop checkpoints, and control mechanisms in a workflow.
- Show how work can be divided into drafting, validating, tailoring, and finalizing roles [file:1].

Classroom / lab exercise

- Final-year emphasis: think in terms of process design, reliability, and handoffs.
- Classroom lab: create a multi-agent resume workflow where one agent drafts, one reviews, one tailors, and one finalizes.
- Students should present the handoff logic and identify where human approval is required.

Session 12

Frameworks, Evaluation, and Responsible AI

What will be covered

- Cover LangChain, LlamaIndex, AutoGen, and CrewAI from an architectural perspective [file:1].
- Discuss workflow vs agent design, evaluation metrics, reliability, bias, fairness, privacy, and responsible AI choices [file:1].
- Close with application-level thinking: what should be built, what should be measured, and what should be governed [file:1].

Classroom / lab exercise

- Final-year emphasis: judge systems by quality, safety, and applicability.
- Classroom lab: present a mini project for an AI resume assistant with architecture, framework choice, metrics, and ethical safeguards.
- Encourage students to explain why their design is practical, measurable, and responsible.